# Rock Mass Rating (RMR89) Calculator

*Geotechnical Classification of Rock Masses Using the Bieniawski (1989) RMR System*

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## Project Links

Live Demo (Streamlit Community Cloud):

*https://rmr89-calculator-n5kcye9bekvh6c6g69j3j7.streamlit.app/*

GitHub Repository:

*https://github.com/aayushkumarlal50-dev/rmr89-calculator*

# Abstract

The Rock Mass Rating (RMR89) Calculator is a production-ready geotechnical engineering application that automates the Bieniawski (1989) Rock Mass Rating classification system. Given field or laboratory measurements of intact rock strength, drill-core quality (RQD), discontinuity spacing, discontinuity condition, and groundwater condition, together with the orientation of the discontinuities relative to the excavation, the tool computes a Final RMR89 score, classifies the rock mass into one of five engineering classes (I-V), and translates that score into practical design parameters: an estimated stand-up time, a Mohr-Coulomb strength range (cohesion and friction angle), basic support guidance, and, where the score permits, an estimate of the Geological Strength Index (GSI).

The tool is deliberately scoped to empirical classification. It does not perform a numerical or limit-equilibrium stability analysis, and it does not replace site-specific structural mapping or laboratory testing. For a given excavation it answers one well-defined question: based on the stated field observations, what is the rock mass quality, and what design parameters does that quality typically imply? This makes it a natural companion to slope-stability and underground-excavation toolkits, supplying the strength and quality inputs that downstream stability analyses require.

# Technical Background

The classification engine implements the six-parameter Bieniawski (1989) RMR system without modification or smoothing. Five parameters - uniaxial compressive strength (UCS) of the intact rock, drill-core quality (RQD), discontinuity spacing, discontinuity condition, and groundwater condition - are each rated independently against the exact step-wise boundaries published in the original 1989 tables. Ratings are looked up against discrete categorical thresholds rather than interpolated linearly between table entries, so that a measured value just either side of a published boundary receives exactly the rating Bieniawski (1989) specifies, with no fractional value invented in between.

The sum of these five ratings gives the Basic RMR, to which a sixth parameter is applied: an orientation adjustment that penalizes discontinuity orientations that are unfavorable to the excavation. Because favorability depends on what is being built, the toolkit maintains three separate orientation-penalty matrices - Tunnels & Mines, Slopes, and Foundations - and applies whichever matrix matches the selected excavation type, reflecting the different kinematic risk that the same discontinuity orientation poses to a tunnel, a slope, or a foundation.

Where the resulting Final RMR89 exceeds 23, the toolkit also reports an estimate of the Geological Strength Index using the widely used Hoek et al. (1995) offset relation, GSI = RMR89 – 5. Below that threshold the correlation is not considered reliable, so the toolkit withholds a GSI estimate rather than report a figure outside the relation's validated range.

# Key Features

- Interactive input engine for real-time processing of UCS, RQD, discontinuity spacing, discontinuity condition, and groundwater condition
- Context-aware orientation penalties, dynamically selected from the Tunnels & Mines, Slopes, or Foundations matrix
- Strict, discrete (non-interpolated) rating boundaries matching the original Bieniawski (1989) tables

- Parameter Contribution Chart - a bar chart illustrating the positive point distribution driving the rock mass's strength
- Quality Degradation Limits chart - a donut chart isolating the parameters penalizing the rock mass
- Automated translation of RMR89 into stand-up time windows, internal friction angle (φ′), cohesion (c′) ranges, and basic support guidelines
- Optional Geological Strength Index (GSI) estimate via the Hoek et al. (1995) offset relation, restricted to RMR89 > 23

# Methodology

## Inputs

- Uniaxial compressive strength (UCS) of the intact rock, or point load strength index
- Drill-core quality (RQD), in percent
- Discontinuity spacing, typically in millimetres
- Discontinuity (joint) condition - roughness, separation/aperture, infilling, weathering, and persistence
- Groundwater condition – general inflow / seepage description or water pressure ratio
- Discontinuity orientation relative to the excavation, and excavation type (Tunnels & Mines / Slopes / Foundations)

## Processing Pipeline

1. Convert each of the five basic parameters into discrete Bieniawski (1989) rating points using strict categorical lookup tables, with no linear interpolation between boundaries.
2. Sum the five basic ratings to obtain the Basic RMR.
3. Apply the orientation adjustment from the excavation-type-specific matrix (Tunnels & Mines, Slopes, or Foundations) to obtain the Final RMR89.
4. Map the Final RMR89 to one of five engineering rock mass classes, from Class I (Very Good) to Class V (Very Poor).
5. Where Final RMR89 > 23, compute GSI = RMR89 – 5; otherwise withhold the GSI estimate.
6. Generate the Parameter Contribution bar chart and Quality Degradation donut chart from the underlying rating breakdown.

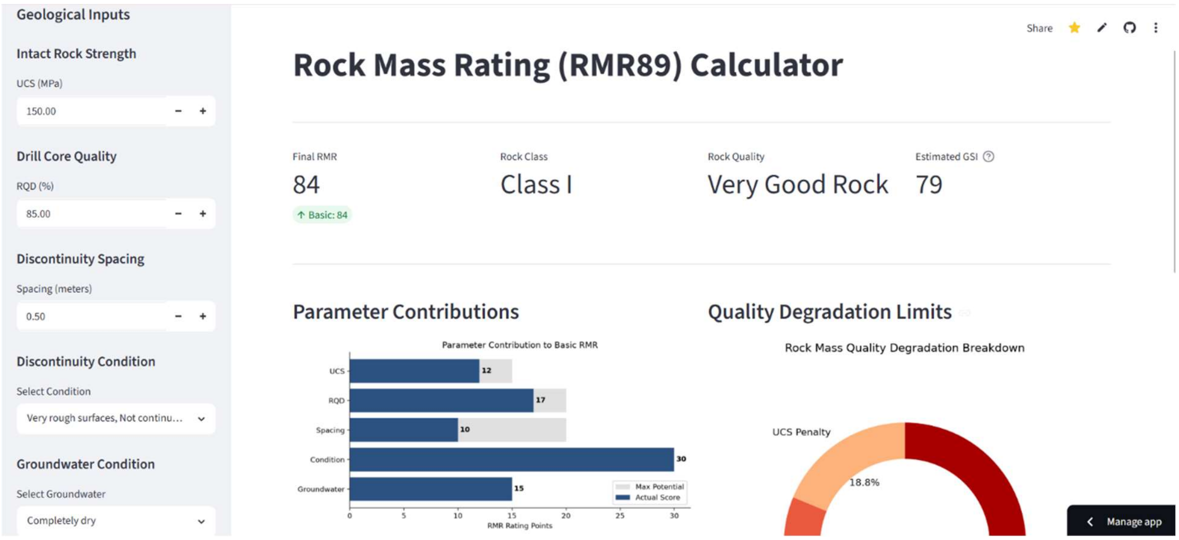
## Outputs

- Final RMR89 score and the underlying point breakdown by parameter
- Rock mass class (I-V) with descriptive quality label
- Estimated stand-up time, internal friction angle (φ′) range, and cohesion (c′) range
- A basic support guideline appropriate to the resulting class
- GSI estimate, when Final RMR89 > 23
- Parameter Contribution and Quality Degradation visualizations

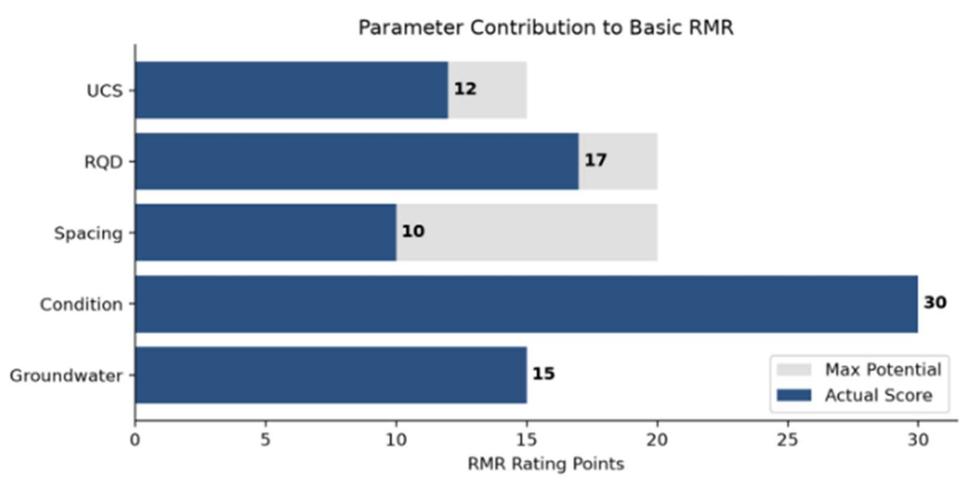
# Screenshots

Screenshots below should illustrate the app interface and a representative result for each major view.

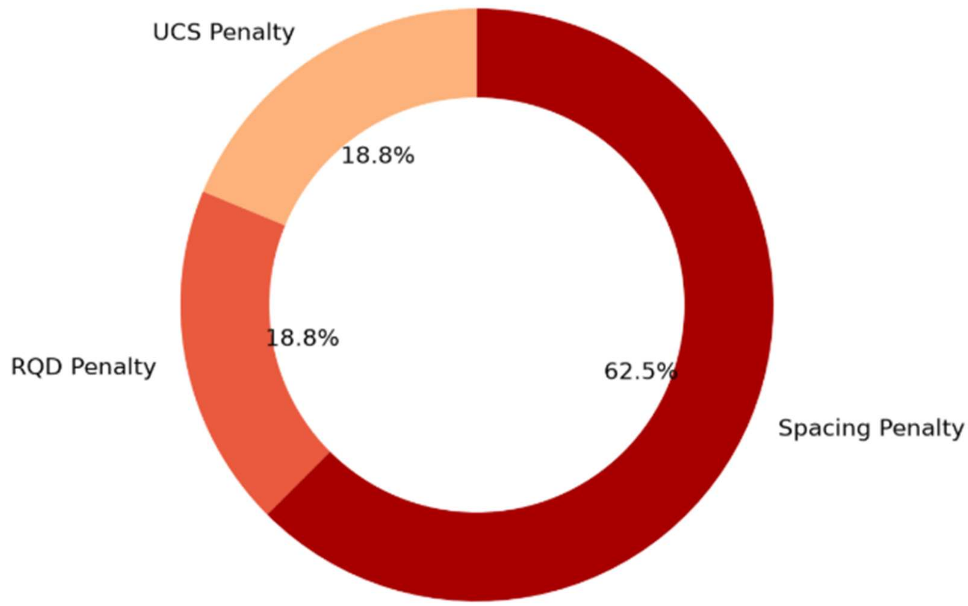
## Input Dashboard




## Parameter Contribution Chart



### Quality Degradation Donut Chart




## System Architecture

### Technology Stack

- Python 3.8+
- Streamlit: interactive web interface
- Matplotlib: chart rendering for the Parameter Contribution and Quality Degradation visualizations
- Pytest: boundary-condition and unit testing

### Codebase Structure

| File | Role |
|---|---|
| `app.py` | Main Streamlit UI and dashboard layout |
| `requirements.txt` | Project dependencies |
| `rmr/constants.py` | Bieniawski reference tables and text definitions |
| `rmr/ratings.py` | Functions converting raw inputs into RMR points |
| `rmr/classification.py` | Mapping Final RMR into engineering classes (I–V) |
| `rmr/calculations.py` | Core engine orchestrating the RMR math |
| `rmr/plotting.py` | Matplotlib visualization generation |
| `tests/` | Pytest suite for boundary and unit testing |

# Assumptions and Scope

This toolkit performs an empirical classification of rock mass quality strictly under the Bieniawski (1989) RMR system. It assumes the input parameters reflect a single, reasonably homogeneous structural domain, and it relies on the user's field judgment for descriptive parameters such as discontinuity condition and groundwater condition. Discrete, non-interpolated rating boundaries are used throughout, consistent with the original 1989 publication, rather than smoothing values across category edges. The orientation adjustment reflects only the excavation-type matrices built into the tool (Tunnels & Mines, Slopes, Foundations) and does not perform an independent kinematic feasibility check of individual discontinuity sets.

The toolkit does not perform numerical stress analysis or limit-equilibrium calculations, does not account for time-dependent rock mass degradation, blast damage, or scale effects, and the GSI estimate is a first-order empirical offset rather than an independent, chart-based GSI evaluation. Results should be treated as a rapid, repeatable classification check, to be confirmed by experienced engineering geological judgment and, where warranted, by further laboratory or in-situ testing before use in final design.

> *Note: An RMR89 classification is a relative quality indicator and an empirical estimator of strength and support parameters - it is not a substitute for site-specific structural mapping, laboratory testing, or numerical / limit-equilibrium design verification.*

# Author and Contact

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